KALAM (கககக) — AGRICULTURAL INSIGHTS ATLAS

Ambasamudram & Cheranmahadevi Taluks, Tirunelveli · Multi-Year EO Analytics (2023–2026)

CROPPING INTENSITY

12.6% Double-Cropped Cropland

CANOPY VIGOUR

95.6% Normal / Vigorous Cropland

ANNUAL WATER DEMAND

348.51 Million m³ (MCM) Gross Requirement

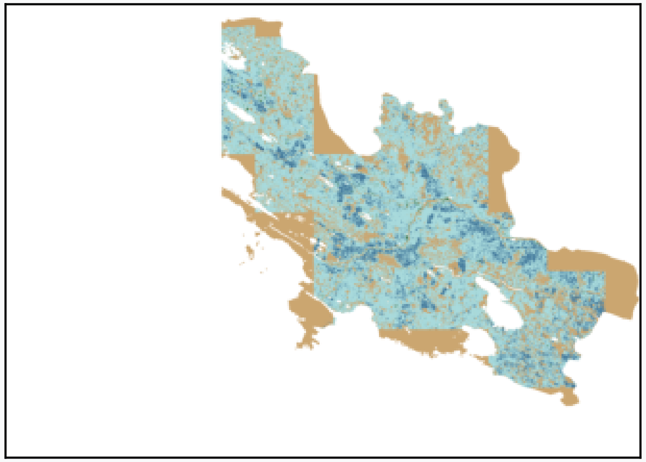
EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

• **Cropping Intensity:**	12.6% of cropland (5288.1 ha) in Ambasamudram & Cheranmahadevi is double-cropped with dual Kar and Pishanam cy...
• **Transplanting Calendar:**	Peak Kar transplanting occurs in May (46.6% of Kar fields), while main Pishanam sowing peaks in Nov (39.5% of ...
• **Irrigation Dependency & Water Access:**	53.8% of paddy acreage (6,440.8 ha) is concentrated within 500 m of river channels, canals, or system irrigati...
• **Crop Change & Farmland Dynamics:**	Evaluated 41,879.6 ha of 200,961.8 ha cropland (confidence ≥ 0.60): 26.3% of paddy and 0.8% of banana parcels ...
• **Crop Health & NDVI Anomaly:**	95.6% of analyzed cropland displays normal or above-normal vegetative vigour. Vegetative moisture/vigour stres...
• **Indicative Gross Irrigation Requirement:**	Total indicative gross irrigation requirement across analyzed farmed plains is 348.5 MCM/year. Paddy cycles (K...
• **07. Groundwater & Recharge Intelligence:**	36.5% of the plains (19,211.0 ha) possess High or Very High groundwater potential. Pumped borewell cropland is...
• **08. Geo-Economic Priority & Action Zoning:**	10,015.2 ha (₹110.3 Cr indicative value at risk) of cropland falls in RESILIENCE, mostly within corridors inun...

Kalam © 2026 · Problem Statement 1 · SustainAI / Mapathon · Contains modified Copernicus Sentinel data

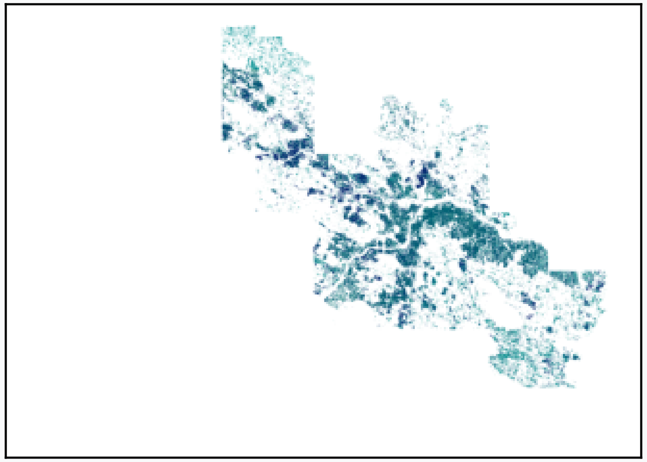
PHENOLOGY DYNAMICS & IRRIGATION ACCESS

01. Cropping Intensity



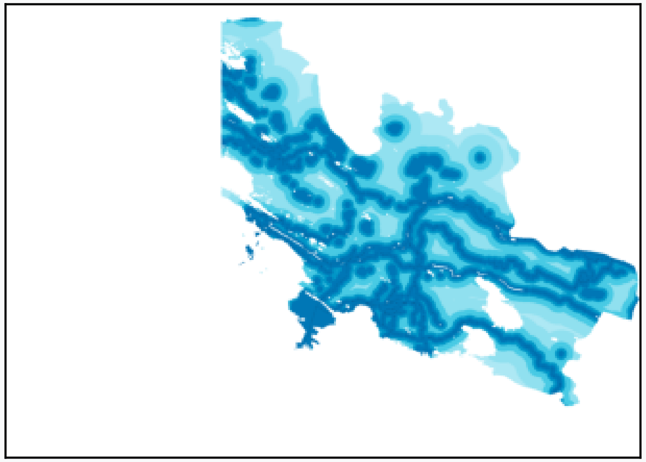
vated (7972.0 ha, 19.0%) | Single Crop (28259.8 ha, 67.5%) | Double Crop (5288.1 ha, 12.6%) | Triple Crop (2.3 ha, 0.0%)

02. Transplanting Calendar



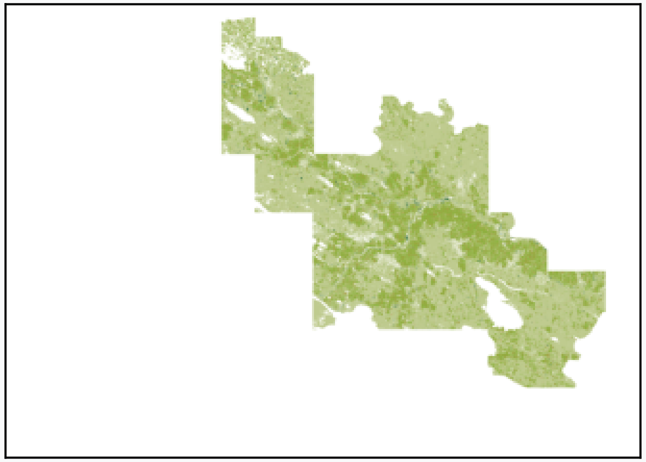
Kar Peak (Jun–Jul) | Kar Late (Aug) | Pishanam Peak (Oct) | Pishanam Late (Nov–Dec)

03. Irrigation Dependency

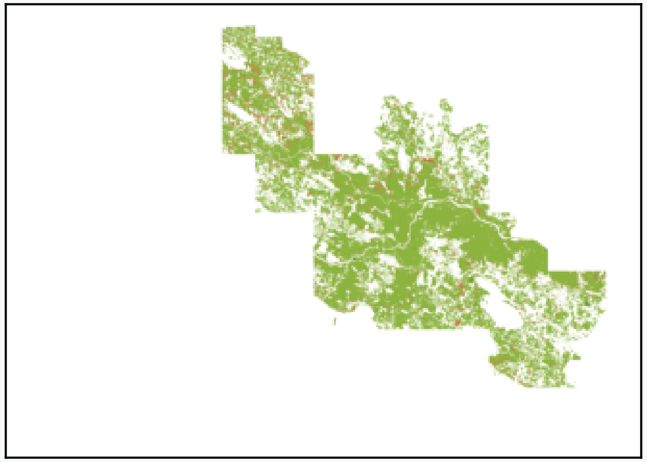


0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

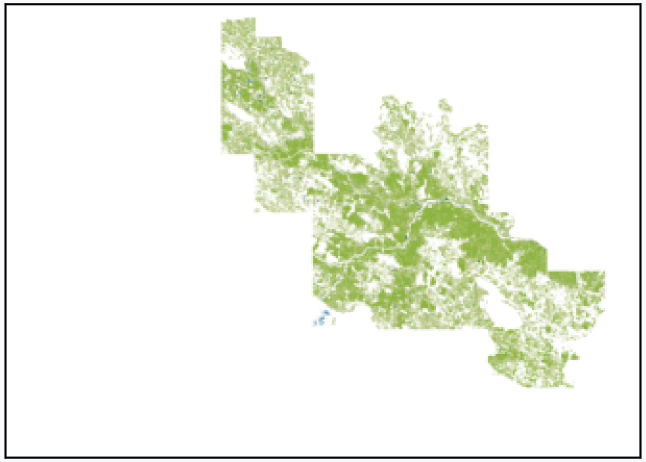
FARMLAND DYNAMICS, VIGOUR & IRRIGATION DEMAND


04. Crop Change & Loss

Stable Paddy | Stable Banana | Candidate Paddy → Banana | Banana → Paddy


05. NDVI Vigour Anomaly

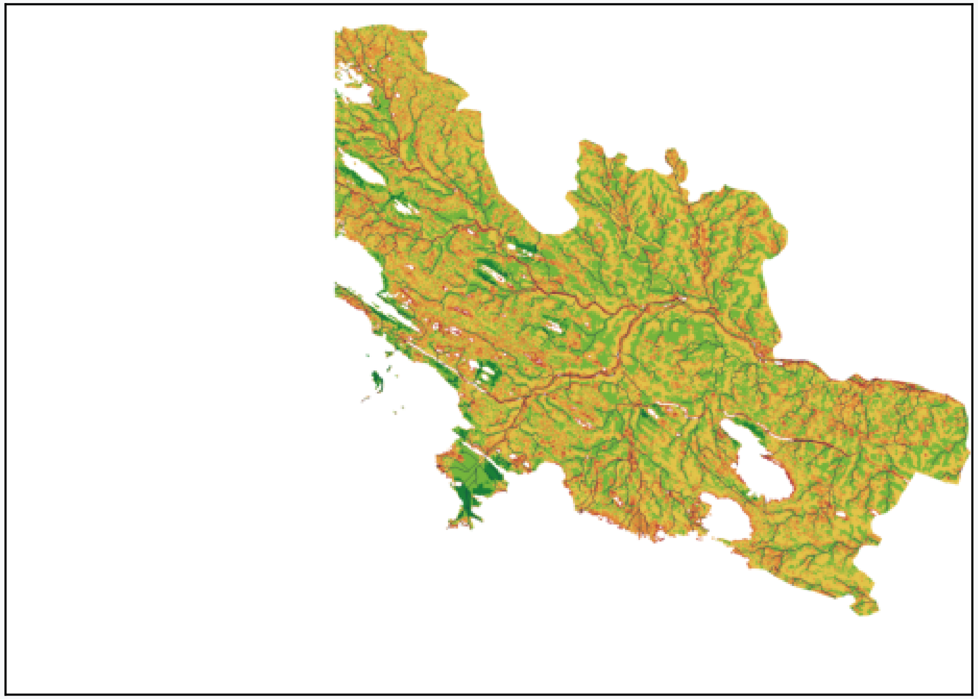
Severe Stress (186.8 ha, 0.7%) | Stressed (914.2 ha, 3.7%) | Normal Vigour (23,153.8 ha, 95.2%) | Above Normal (93.0 ha, 0.4%)


06. Water Demand Budget

Kar Paddy (Short-duration) (117.76 MCM, 33.8%) | Pishanam Paddy (Main Monsoon) (143.61 MCM, 41.2%) | Perennial Banana (Year-round) (7.4 MCM, 2.1%

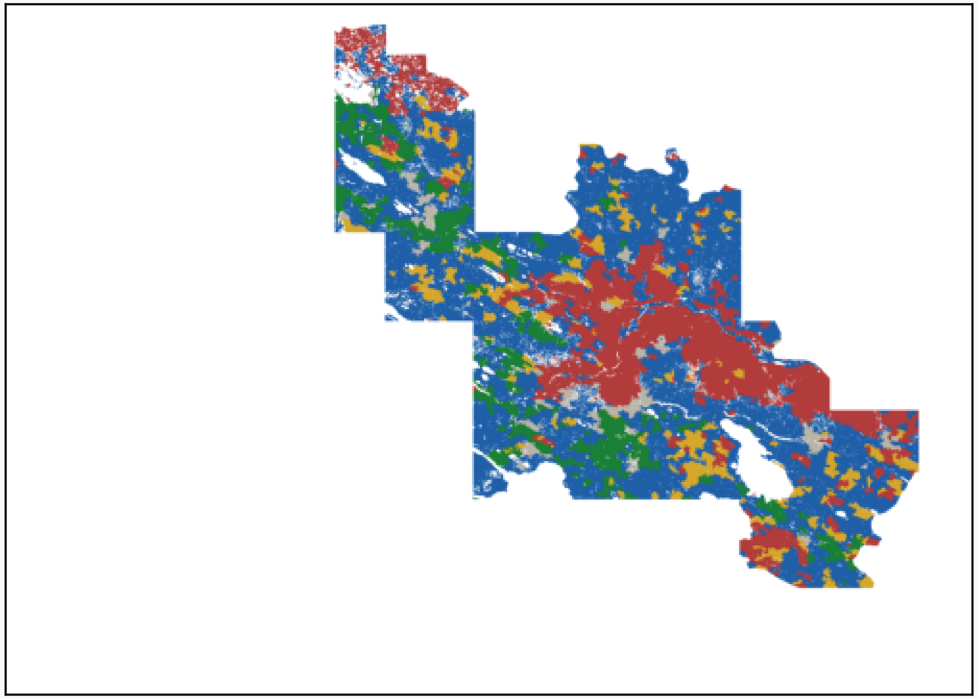
HYDROGEOLOGY & GEO-ECONOMIC INTERVENTION ZONING

07. Groundwater Potential & MAR Siting



Very Low Potential | Low Potential | Moderate Potential | High Potential | Very High Potential

08. Geo-Economic Action Priority



RESILIENCE | PROTECT | OPTIMIZE | INVEST | Stable / monitor